

Smart stethoscope

Analyzing heart and lungs



Problem statement



Heart and lung illnesses are rising in India, driven by a combination of worsening air pollution, lifestyle changes, stress, and poor access to optimal healthcare. These illnesses are typically chronic and genetically predisposed.



In a health emergency, patients may be unable to reach the hospital due to financial barriers, travel distance, or lack of transportation. They may also not recognize the severity and delaying needed care.



There is also a severe shortage of doctors and qualified healthcare professionals in rural India, thus residents receive partial or no care for serious conditions, leading to poor health outcomes.



As traditional stethoscopes allow doctors to listen to heart and lung sounds only in real time, they lack the capability to record, analyze, or share data for further review.

Solution

Our project lets doctors remotely monitor heart and lung patients with minimal contact. Patients can track their vitals at home, with real-time data sent via WiFi to doctors. The system detects irregularities and sends alerts, improving accessibility, efficiency, and convenience for all.



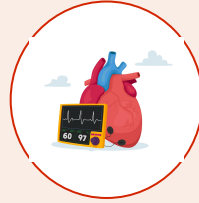
Objective



**Digitally
capture heart
and lung
sounds**



**Utilize
arduino for
signal
processing**



**Analyze
sounds for
abnormalities**



**Store and
share data
securely**



**Facilitate
healthcare
support**

Components

1

Stethoscope

2

Arduino uno

3

Microphone

4

Amplifier circuit

5

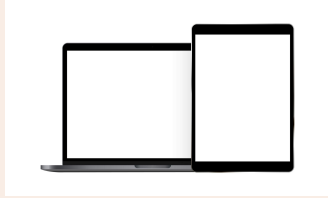
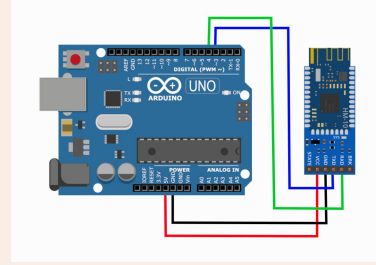
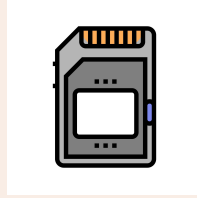
Sd card

6

Bluetooth / wifi module

7

Mobile / laptop



application

Detection of cardiac abnormalities: identify heart murmurs, arrhythmias, and other irregularities.

Lung condition monitoring: track respiratory issues such as asthma, pneumonia, and other lung infections.

Telemedicine support: enable remote consultations by sharing real time or record heart and lung sounds.

Medical training and education: serve as a learning tool for students to study real physiological sounds.

Rural and remote healthcare: enable diagnostic support in areas with limited access to medical facilities.

conclusion

The Smart Stethoscope using Arduino bridges the gap between traditional auscultation and modern digital healthcare. It enables doctors to record, analyze, and share patient data, providing better diagnostic accuracy and remote healthcare solutions.